## 2. Exercise

# Linear models and regression

## Exercise Sheet 2

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**Solve the following problems before the exercise class on November 5.**

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**Problem 1**
In an experiment, the values 7.8, 13.5, 7.5, 8.1 and 10.6 are measured. It can be assumed that they are independent realisations of a normal distribution with unknown expectation $\mu \in \mathbb{R}$ and known standard deviation $\sigma = 3$.

a) Compute a confidence interval for $\mu$ with error level 0.1.

b) The goal of the experiment is to verify that $\mu$ is greater than 8.5. Formulate the null hypothesis and the alternative and perform a statistical test with level 0.05.

*Hint: The standard normal distribution has the following quantiles:*

| $p$ | 0.9 | 0.95 | 0.975 | 0.99 | 0.995 |
|---|---|---|---|---|---|
| $z_p$ | 1.28 | 1.64 | 1.96 | 2.33 | 2.58 |

**Problem 2**
Let $n, m \in \mathbb{N}$ and let $\sigma_1, \sigma_2 \in (0, \infty)$ be fixed. For $\mu_1, \mu_2 \in \mathbb{R}$ let $X_1, \dots, X_n \sim \text{Normal}(\mu_1, \sigma_1^2)$ and $Y_1, \dots, Y_m \sim \text{Normal}(\mu_2, \sigma_2^2)$ be all independent. For unknown $\mu_1, \mu_2 \in \mathbb{R}$, realizations of the random vectors $\mathbf{X} = (X_1, \dots, X_n)^T$ and $\mathbf{Y} = (Y_1, \dots, Y_m)^T$ are given. Recall that

$$\overline{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{and} \quad \overline{\mathbf{Y}} = \frac{1}{m}\sum_{i=1}^{m} Y_i.$$

a) One is interested in the test problem

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2.$$

Show that, for $\alpha \in (0, 1)$,

$$\varphi_{=}(\mathbf{X}, \mathbf{Y}) = \begin{cases} \text{cannot reject}, & T \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}), \\ \text{reject}, & T \notin (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}), \end{cases} \quad \text{with} \quad T = \frac{\overline{\mathbf{X}} - \overline{\mathbf{Y}}}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}}$$

is a level-$\alpha$ test for $H_0$.

b) Provide a level-$\alpha$ test for

$$H_0 : \mu_1 \leq \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2.$$

## Confidence Interval

$$\left(\mathcal{X}, \begin{matrix} P(\mathcal{X}) \\ \mathcal{B}(\mathcal{X}) \end{matrix}, (\mathbb{P}_\theta)_{\theta \in \Theta}\right) \qquad \underline{g(\theta)}\ ?$$

estimator

$I_x$ is a confidence region for $g(\theta)$ with error level $\alpha \in (0,1)$

if $\mathbb{P}_\theta(g(\theta) \in I_x)$

Eg: $\mathcal{N}(\mu_p, \sigma_0^2)$ → known

$\mu_p$ → unknown

$$I_x = \left(\bar{x}_n - z_{1-\frac{\alpha}{2}} \frac{\sigma_0}{\sqrt{n}}, \bar{x}_n + z_{1-\frac{\alpha}{2}} \frac{\sigma_0}{\sqrt{n}}\right)$$

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## #Problem 1

a) We have $n = 5$, $\alpha = 0.1$

$$\bar{x}_5 = \frac{1}{5}(7.8 + 13.5 + 7.5 + 8.1 + 10.6)$$
$$= 9.5$$

$$z_{1-\frac{\alpha}{2}} = z_{0.95} = 1.64$$

$$I_x = \left(9.5 - 1.64 \cdot \frac{3}{\sqrt{5}}, 9.5 + 1.64 \frac{3}{\sqrt{5}}\right)$$

$= (7.2997, 11.70029)$

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## Hypothesis testing

$H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$

Eg: $\mathcal{N}(\mu, \sigma_0^2)$

$\mu$ → unknown

$\sigma_0^2$ → known

$H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$

$$\varphi : x \longmapsto \begin{cases} \text{cannot reject} & \bar{x} < \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}} \\ \text{reject} & \bar{x} \geq \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}} \end{cases}$$

$H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$

$$\varphi : x \longmapsto \begin{cases} \text{cannot reject} & \bar{x} \in \left(\mu_0 - z_{1-\frac{\alpha}{2}} \frac{\sigma_0^2}{\sqrt{n}}, \mu_0 + z_{1-\frac{\alpha}{2}} \frac{\sigma_0^2}{\sqrt{n}}\right) \\ \text{reject} & \longrightarrow \bar{x} \notin (\text{"}, \text{"}) \end{cases}$$

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b) Hypothesis, $H_0 : \mu \leq 8.5$ vs $H_1 : \mu > 8.5$

We have $\alpha = 0.05$, $\mu = 8.5$ $z_{1-\alpha} = z_{0.95} = 1.64$

$$\mu_0 + z_{1-\alpha}\frac{\sigma_0}{\sqrt{n}} = 8.5 + 1.64 \cdot \frac{3}{\sqrt{5}} = 10.70029$$

$$\bar{x}_5 = 9.5$$

We have $\bar{x}_5 < \mu_0 + z_{1-\alpha}\frac{\sigma}{\sqrt{n}}$

$\therefore$ We cannot <u>reject</u> $H_0$ i.e $\mu \leq 8.5$

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$$H_0 : \theta \in \Theta_0 \quad \text{vs} \quad H_1 : \theta \in \Theta_1$$

$$\varphi : x \longrightarrow \{\text{cannot reject, reject}\}$$

$\varphi$ is a level $\alpha$ test if

$$\mathbb{P}_\theta(\varphi(x) = \text{reject}) \leq \alpha \quad \forall \theta \in \Theta_0$$

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# Problem 2

a) $H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$

$\Leftrightarrow H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs} \quad H_1 : \mu_1 - \mu_2 \neq 0$

$$\varphi(X, Y) = \begin{cases} \text{cannot reject} \end{cases}$$

$\varphi_{=}(\cdot,\cdot) \begin{cases} \\ \\ \end{cases}$ reject

Let $\mu_1, \mu_2 \in \mathbb{R}$ s.t. $\mu_1 - \mu_2 = 0$

we have

$$\mathbb{P}_{\mu_1,\mu_2}(\varphi(X,Y) = \text{reject})$$

$$= \mathbb{P}_{\mu_1,\mu_2}\left(T \notin \left(-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}\right)\right)$$

$$= 1 - \mathbb{P}_{\mu_1,\mu_2}\left(T \in \left(-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}\right)\right)$$

$$= 1 - \mathbb{P}_{\mu_1,\mu_2}\left(-z_{1-\frac{\alpha}{2}} \leq T \leq z_{1-\frac{\alpha}{2}}\right)$$

$\hookrightarrow <$

$$= 1 - \left(\mathbb{P}\left(T \leq z_{1-\frac{\alpha}{2}}\right) - \mathbb{P}\left(T \leq -z_{1-\frac{\alpha}{2}}\right)\right)$$

#Recall

If $X$ and r.v $P(X=a)=0$

$\mathbb{P}(a < X \leq b) = F(b) - F(a)$

If I have

$X_i \sim \mathcal{N}(\mu, \sigma^2)$

then

$\frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$

$N_1(\mu_1, \sigma_1^2)$ & $N_2(\mu_2, \sigma_2^2)$

$\Downarrow$

$N_1 \perp\!\!\!\perp N_2$

independent

$\hookrightarrow N_1 + N_2 = \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

Let us find the distributions of T

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}}, \text{ with}$$

$$\bar{X} \sim \mathcal{N}\left(\mu_1, \frac{\sigma_1^2}{n}\right), \quad \bar{Y} \sim \mathcal{N}\left(\mu_2, \frac{\sigma_2^2}{m}\right)$$

$$\therefore \quad \bar{X} - \bar{Y} \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}\right)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0,1) \rightarrow \text{standard normal}$$

Recall
if $X \sim \mathcal{N}(\mu, \sigma^2)$
$\Rightarrow \frac{X - \mu}{\sigma} \sim \mathcal{N}(0,1)$

Therefore

$$\mathbb{P}_{\mu_1, \mu_2}(\varphi(X, Y) = \text{reject})$$

$$= 1 - \left(1 - \frac{\alpha}{2} - \left(1 - \left(1 - \frac{\alpha}{2}\right)\right)\right)$$

$$= \alpha \leq \alpha$$

hence it is level-$\alpha$ test

by quantile
$P(X \leq q_p) = p$
$P(X \leq -q_p) = 1-p$

b) level-$\alpha$ test for $H_0: \mu_1 \leq \mu_2$ vs $H_1: \mu_1 > \mu_2$

$$\mathbb{Q}_{\leq}(X,Y) = \begin{cases} \text{cannot reject} & T \in (-\infty, z_{1-\alpha}) \\ \text{reject} & T \notin (-\infty, z_{1-\alpha}) \end{cases}$$

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# Problem 3

a) We first find the p.df of $X_1$

$$f_\alpha(u) = F_\alpha'(u)$$

$$= \begin{cases} \alpha u^{-\alpha-1} & u \geq 1 \\ 0 & u < 1 \end{cases}$$

The likelihood function is given by

$$\ell(\alpha) = \prod_{i=1}^{n} f_\alpha(x_i) \quad \text{for } x_i \geq 1$$

$$= \prod_{i=1}^{n} \alpha \cdot x_i^{-\alpha-1}$$

$$= \alpha^n \prod_{i=1}^{n} x_i^{-\alpha-1}$$

The log likelihood function is

$$L(\alpha) = \log(l(\alpha))$$

$$= n \log \alpha + \sum_{i=1}^{n} (-\alpha - 1) \log x_i$$

$$= n \log \alpha - (\alpha + 1) \sum_{i=1}^{n} \log x_i$$

We have,

$$\frac{d}{d\alpha} L(\alpha) = \frac{n}{\alpha} - \sum_{i=1}^{n} \log x_i$$

We solve $\frac{d}{d\alpha} L(\alpha) = 0$ and obtains

$$\frac{n}{\alpha} - \sum_{i=1}^{n} \log x_i \iff \hat{\alpha} = \frac{n}{\sum_{i=1}^{n} \log x_i}$$

Also $\frac{d^2}{d\alpha^2} L(\alpha) = \frac{-n}{\alpha^2}$

$$\Rightarrow \frac{d^2}{d\alpha^2} L(\hat{\alpha}) = -\frac{n}{\hat{\alpha}^2} \leq 0$$

$$\Downarrow$$

Thus

$$\underline{\underline{\hat{\alpha} = \hat{\alpha}_{MLE}}}$$

b) $$\hat{\alpha} = \frac{n}{\sum_{i=1}^{n} \log x_i} = \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \log x_i}$$

Recall $$\frac{1}{n}\sum x_i \xrightarrow{\text{P.a.s}} \mathbb{E}[X_1]$$

We have

$$E[\log X_1] = \int_1^{\infty} \log u \cdot f_\alpha(u)\, du$$

$$= \alpha \int_1^{\infty} u^{-\alpha-1} \log u \, du$$

$$u' = u^{-\alpha-1} \longrightarrow u = -\frac{1}{\alpha} u^{-\alpha}$$

$$v = \log u \longrightarrow v' = \frac{1}{u}$$

Integration by parts

$$E[\log X_1] = \alpha \left[ \left[ \overbrace{-\frac{1}{\alpha} u^{-\alpha} \log u}^{0} \right]_1^{\infty} + \frac{1}{\alpha} \int_1^{\infty} u^{-\alpha-1} du \right]$$

$$= \cancel{\alpha} \frac{1}{\cancel{\alpha}} \left[ -\frac{1}{\alpha} u^{-\alpha} \right]_1^{\infty}$$

$$= \frac{1}{\alpha}$$

From the SLLN & WLLN

$$\frac{1}{n}\sum_{i=1}^{n} \log x_i \xrightarrow{a.s,\ \mathbb{P}} E[\log x_i] = \frac{1}{\alpha}$$

$$\Rightarrow \quad \hat{\alpha} \xrightarrow{\mathbb{P},\ a.s} \alpha$$

i.e., $\hat{\alpha}$ is strongly & weakly consistent

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# Problem 4

a) The Likelihood function

$$\ell(p,q) = \prod_{i=1}^{n} \mathbb{P}_{p,q}(x_i) \qquad x_i \in \{0,1,2\}$$

$$= (1-p-q)^{n-N_1-N_2}\, p^{N_1} p^{N_2}$$

$$\Rightarrow L(p,q) = \log(\ell(p,q))$$

$$= (n-N_1-N_2)\log(1-p-q) + N_1 \log p + N_2 \log q$$

$$\frac{\partial}{\partial p} L(p,q) = -\frac{n-N_1-N_2}{1-p-q} + \frac{N_1}{p}$$

$$\frac{\partial}{\partial q} L(p,q) = \frac{-n-N_1-N_2}{1-p-q} + \frac{N_2}{q}$$

We solve $\begin{cases} \frac{\partial}{\partial p} L(p,q) = 0 \\ \frac{\partial}{\partial q} L(p,q) = 0 \end{cases}$

$$\Leftrightarrow \begin{cases} \frac{N_1}{p} - \frac{N_2}{q} = 0 \\ \frac{-n-N_1-N_2}{1-p-q} + \frac{N_2}{q} = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} N_1 q - N_2 p = 0 \\ (-n-N_1-N_2)q + N_2(1-p-q) = 0 \end{cases}$$

By solving above

$$\Leftrightarrow \quad \hat{p} = \frac{N_1}{n} \qquad \hat{q} = \frac{N_2}{n}$$

$$\text{Hessian}(p,q) = \begin{pmatrix} \frac{\partial^2}{\partial p^2} L & \frac{\partial^2}{\partial p \partial q} L \\ \frac{\partial^2}{\partial p \partial q} L & \frac{\partial^2}{\partial q^2} L \end{pmatrix}$$

$$\det \mathrm{Hess}(\hat{p}, \hat{q}) > 0 \qquad \hat{p} = p_{MLE}, \ \hat{q} = q_{MLE}$$

$$\Rightarrow \frac{\partial^2}{\partial p^2} L(\hat{p}, \hat{q}) < 0$$

b) $$\mathbb{E}[\hat{p}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{x_i = 1\}}\right]$$

$$= \frac{1}{n}\sum \underbrace{\mathbb{P}(x_i = 1)}_{p}$$

$$= \frac{1}{n} n p$$

$$= p$$

and similarly

$$\mathbb{E}[\hat{q}] = q$$

$\therefore (\hat{p}, \hat{q})^T$ is unbiased estimator $(p, q)$